explain code splitting and lazy loading in erformance Optimization in React ,,,why to use them what is purpose ,which new terms use etc explain in easy wording roman m explain kro jasy pehli dfa smjh rhi hu mn

React mein **performance optimization** ka matlab hota hai app ko fast, smooth aur light banana. Is mein 2 important concepts hotay hain:

1. Code Splitting kya hota hai?

Code splitting ka simple matlab:

Tumhara React app ka poora code ek hi baar load na ho, balkay **chhoti chhoti files (chunks)** mein toot jaye aur zarurat par load ho.

Simple example:

Socho tumhari app ek bari kitab hai 📘
Agar start mein poori kitab ek sath de di jaye to slow ho jati hai.

Code splitting mein:

- sirf pehla chapter start mein aata hai
- baqi chapters tab load hotay hain jab zarurat ho

Why use it?

- App fast load hoti hai
- Initial loading time kam hota hai
- User ko jaldi UI nazar aata hai

---

2. Lazy Loading kya hota hai?

Lazy loading ka matlab:

👉 kisi cheez ko **tab load karna jab wo actually chahiye ho**

React mein usually components ko lazy load kiya jata hai.

Example:

Tumhare app mein 3 pages hain:

- Home
- About
- Profile

Agar user Home par hai, to:

- About aur Profile abhi load na hon
- wo tab load hon jab user un par click kare

React mein kaise hota hai?

React mein 2 cheezen use hoti hain:

1. `React.lazy()`

Is se component lazy load hota hai

```
```

JavaScript

Matlab:
👉 About component ab tab load hoga jab zarurat hogi

---

2. `Suspense`

Jab component load ho raha ho to loading show karne ke liye

```
<Suspense fallback={<div>Loading...</div>}>
```

JavaScript

Code Splitting vs Lazy Loading difference

Code Splitting:

- Poore app ko chunks mein divide karna

Lazy Loading:

- Specific component ko tab load karna jab zarurat ho

👉 Simple relation:
Lazy loading = code splitting ka ek part hai

Kyun use karte hain? (Important points)

- App fast open hota hai
- Unnecessary code load nahi hota
- Data aur memory save hoti hai
- User experience better hota hai
- Large apps ko manageable banata hai

New terms jo tumhe milain ge

- **Chunks** → small parts of code
- **Bundle** → combined JS file
- **Dynamic Import** → `import()` at runtime
- **Suspense fallback** → loading UI
- **Route-based splitting** → pages ke basis par splitting

Agar tum chaho to main tumhe iska **real React project example (routing wala)** bhi step by step bana ke samjha sakta hoon 👍

also code b dw

Theek hai 👍 ab main tumhe **complete simple React example** de raha hoon jisme **code splitting + lazy loading + routing** sab clear ho jaye.

🚀 React Example (Lazy Loading + Code Splitting)

📁 1. App.js (Main File)

```
import React, { Suspense } from "react";
import { BrowserRouter as Router, Routes, Route, Link } from "react-router-dom";

// Lazy loading components
const Home = React.lazy(() => import("./Home"));
const About = React.lazy(() => import("./About"));
const Profile = React.lazy(() => import("./Profile"));

function App() {
  return (
    <Router>
      <nav>
        <Link to="/">Home</Link> |{" "}
        <Link to="/about">About</Link> |{" "}
        <Link to="/profile">Profile</Link>
      </nav>
```

JavaScript

📁 2. Home.js

```
import React from "react";

function Home() {
  return <h1>This is Home Page</h1>;
}
```

JavaScript

📁 3. About.js

```javascript
import React from "react";

function About() {
  return <h1>This is About Page</h1>;
```

JavaScript

📁 4. Profile.js

```javascript
import React from "react";

function Profile() {
  return <h1>This is Profile Page</h1>;
}
```

JavaScript

🧠 Ab samajh lo flow (simple roman urdu)

🔷 Step 1:

App start hoti hai → sirf **App.js load hota hai**

🔷 Step 2:

Home, About, Profile **abhi load nahi hotay**

🔷 Step 3:

Jab tum click karti ho:

- About pe → tab About.js load hota hai
- Profile pe → tab Profile.js load hota hai

🔷 Step 4:

Jab tak load ho raha hota hai:

👉 "Loading page..." show hota hai (Suspense fallback)

⚡ Final Simple Idea

- **Code Splitting** → app ko parts mein todna
- **Lazy Loading** → parts ko zarurat par load karna
- **Suspense** → loading screen show karna

Agar tum chaho to main tumhe next step bhi sikha sakta hoon:

👉 "↳ React app ko aur fast kaise banate hain (memo, useMemo, useCallback)"

Sources

react.lazy and suspenese etc explain

Theek hai 👍 ab main tumhe **React.lazy() aur Suspense** bilkul simple roman urdu mein samjhata hoon, jese first time samajh rahi ho.

⚡ 1. React.lazy() kya hota hai?

👉 Simple meaning:

`React.lazy()` ka kaam hai **component ko tab load karna jab uski zarurat ho**.

Normal React mein:

- saare components pehle hi load ho jate hain ❌ (slow)

Lazy loading mein:

- sirf woh component load hota hai jab user us page par jaye ✅ (fast)

📌 Syntax:

JavaScript

🧠 Iska simple matlab:

👉 "About component ko tab load karo jab user usay open kare"

📦 Without lazy:

- sab pages pehle se load
- app slow

⚡ With lazy:

- sirf current page load
- baqi baad mein

⏳ 2. Suspense kya hota hai?

👉 Simple meaning:

`Suspense` ek wrapper hota hai jo **loading state handle karta hai**

Jab lazy component load ho raha hota hai:
👉 Suspense ek "backup UI" show karta hai (loading screen)

📌 Syntax:

```javascript
<Suspense fallback={<h1>Loading...</h1>}>
```

JavaScript

🧠 Simple explanation:

- jab tak About load ho raha hai
  👉 “Loading…” show hoga
- jab load ho jaye
  👉 actual page show ho jayega

🔥 Real life example:

Socho tum YouTube open karti ho:

- Video abhi load ho raha hai ⏳
- Screen pe “buffering…” aa raha hai
- Video ready → play start 🎥

👉 buffering = Suspense fallback
👉 video load = React.lazy

🧩 Dono ka relation

Concept	Kaam
React.lazy	component ko delay se load karta hai
Suspense	loading time pe UI show karta hai

👉 Dono mil kar kaam karte hain

🚀 Full mini example:

```
import React, { Suspense } from "react";

const About = React.lazy(() => import("./About"));
```



```
    <Suspense fallback={<h2>Loading...</h2>}>
      <About />
    </Suspense>
```

JavaScript

```javascript
export default App;
```

⚡ Why use them?

- App fast hota hai 🚀
- unnecessary code load nahi hota
- user ko smooth experience milta hai
- large apps manageable hoti hain

Agar tum chaho to main next step mein tumhe ↪ **real project routing + lazy loading with interview questions** bhi sikha sakta hoon 👍

Sources